

MAINTENANCE ANALYTICS — CASE STUDY

Mapping Unscheduled Maintenance Events Across the Airbus In-Production Fleet

A data-driven analysis of FAA Service Difficulty Reports (2015–2025) identifying where maintenance events concentrate by aircraft system, how failure patterns shift with aircraft age, and what recurring themes emerge from technician narratives — with recommendations for maintenance planning and reporting.

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Data source FAA Service Difficulty Reports, 2015–2025 (US-registered operators)

Live dashboard airbus-maintenance-analytics-igorlaminsi.streamlit.app

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Executive Summary

This report analyzes 112,195 FAA Service Difficulty Reports filed between 2015 and 2025 on Airbus A320, A330 and A350 aircraft operated by US-based carriers. The goal is to answer a maintenance-planning question in plain terms: **which aircraft systems generate the most unscheduled maintenance activity, does that activity change with aircraft age, and what does the underlying technical language tell us that the counts alone don't?**

112,195

reports analyzed
(A320 / A330 / A350,
2015–2025)

2,819

distinct aircraft
observed
across the three
families

30%

of all reports
concentrate
in a single ATA chapter
(Fuselage)

7 / 46

ATA chapters account
for
82% of report volume

Key findings

- **Maintenance volume is highly concentrated, not evenly spread.** Seven of forty-six ATA chapters — Fuselage, Lights, Doors, Equipment/Furnishings, Air Conditioning, Stabilizers and Wings — account for 82% of all reports. Fuselage alone represents 30%.
- **Failure patterns split cleanly into two profiles by aircraft age.** Structural systems (Fuselage, Wings) are increasingly reported as aircraft age — consistent with wear and fatigue. Cabin and system-level equipment (Air Conditioning, Engine, Doors) are reported more on younger aircraft — consistent with early-life / entry-into-service defects.
- **Technician narratives confirm and explain the age pattern.** "Corrosion" is the most distinctive term for both Fuselage and Wings — a textbook wear signature. Air Conditioning reports, by contrast, are dominated by "odor" and "smell," not mechanical failure vocabulary — a different kind of problem entirely.
- **The raw upward trend in report volume is mostly a fleet-growth story.** Reports rose $\times 2.5$ from 2015 to 2025, but the observed fleet grew $\times 1.9$ over the same window. Once normalized per 1,000 aircraft, the trend is flat with a clear dip in 2020 (COVID-19 reduction in operations).

Why this matters for maintenance planning

These four findings, taken together, point to different management levers: a *volume/workload* problem concentrated in a handful of systems, an *age-based inspection* problem for structural fatigue, an *early-life quality* problem for younger aircraft, and a *reporting-vs-reality* problem when reading raw trend counts. Section 11 turns each of these into a concrete action.

Project Overview

Airlines and MRO (Maintenance, Repair & Overhaul) organizations generate large volumes of unscheduled maintenance activity — repairs, replacements, and inspections triggered by a fault found in service rather than by a planned schedule. Each event is costly in isolation, but the real planning value comes from looking at the pattern across thousands of events: which aircraft systems create the most recurring work, whether that work changes as aircraft age, and what technicians are actually describing when they write up a fault.

This project builds that pattern from public failure-report data on the current Airbus production fleet (A320, A330, A350). It is structured as four linked analyses:

Analysis	Question answered
ATA mapping	Where does report volume concentrate, by aircraft system (ATA chapter)? Which systems matter most once normalized by fleet size?
Age & wear	Do specific systems get reported more as an aircraft ages (wear/fatigue), or less (early-life defects)?
Text mining	What do technicians actually write in fault descriptions? What terms are distinctive to each system?
Temporal trend	Is unscheduled maintenance activity increasing over time — and if so, is that a reliability signal or a fleet-growth artifact?

The output is a set of documented findings (this report and an interactive dashboard) designed to support two different readers: an operations decision-maker who needs the "so what" in plain language, and a data reviewer who needs to see the method behind every number.

Scope of the underlying question

This is a portfolio case study built on public data, not an internal engineering audit. Its value is in demonstrating a repeatable analytical method — one that a maintenance analytics team could apply directly to internal, higher-quality operational data (see Section 12).

Data & Scope

Source

[FAA Service Difficulty Reports \(SDR\)](#) — the US Federal Aviation Administration's public register of aircraft discrepancies, failures and defects, submitted by airlines and maintenance stations. Twelve annual files (2015–2026) were combined; 2026 is excluded from year-over-year comparisons as it is a partial year at the time of analysis.

Fleet families in scope

The analysis covers the Airbus families currently in production and active commercial service: **A320** (including A318/A319/A320/A321, ceo and neo) **A330** **A350** .

Family	Reports (2015–2025)	Decision	Reason
A320	97,794	In scope	—
A300	16,636	Excluded	Earlier generation, production ended 2007
A330	13,742	In scope	—
A220	2,903	Excluded	Originally a Bombardier design (C Series), not native Airbus engineering
A350	659	In scope (small-sample caveat)	—
A310	186	Excluded	Earlier generation, production ended 1998
Helicopters	20	Excluded	Different aircraft category
A380	19	Excluded	Production ended 2021, negligible sample
A340	1	Excluded	A330-era design, production ended ~2011

Resulting analytical dataset

112,195

reports in scope
(complete years,
A320/A330/A350)

2,568

distinct A320-family
aircraft observed

210

distinct A330
aircraft observed

41

distinct A350
aircraft observed

"Distinct aircraft observed" counts unique serial numbers that appear in at least one report — see Section 04 for why this is a lower bound on true fleet size, not a fleet census.

Assumptions

Every analytical result rests on choices. They are stated here explicitly so each number in this report can be traced back to a decision, not treated as a black box.

Geographic coverage

SDR data only covers aircraft operated by **US-based operators**. Every result in this report is a proxy for the Airbus fleet operating in the United States — **not** a representative view of the global Airbus fleet. This bias is assumed and reported deliberately, not hidden.

Reporting bias

A report only exists if a problem was noticed *and* filed. Two consequences follow:

- Report counts reflect what was **reported**, not necessarily everything that occurred.
- "Distinct aircraft observed" — used as a fleet-size proxy to normalize comparisons — **understates true fleet size**: an aircraft with zero reports never appears in this data, so it is never counted.

Aircraft age proxy

Aircraft age is measured in **flight cycles** rather than calendar age, and age bands (young / mid-life / aged) are computed as **terciles within each family** rather than a single global threshold. A long-haul A350 accumulates cycles far more slowly than a short-haul A320; a single global cutoff would misclassify almost every A350 as "young." A small number of clearly erroneous cycle/hour values (up to 5.68 million cycles for a single aircraft) were nulled rather than capped, to avoid disguising bad data as a genuinely old aircraft.

Undocumented fields

The `NatureOfConditionA` field (a letter code for fault nature) has no publicly available definition — confirmed after retrieving and reading the FAA's official "SDRS Field Instructions" document, which references an internal dropdown list rather than a published table. These codes are retained in the data but never used as a basis for a conclusion in this report; the free-text fault descriptions (Section 08) are used instead.

Methodology

Every method in this report was chosen to stay explainable: each metric can be described in a sentence and reproduced from the raw data. No black-box modeling is used for the headline findings.

Classification & cleaning

Aircraft model codes were mapped to families (A320/A330/A350) via an explicit prefix rule, verified line-by-line against known reference counts. Records outside the three families are flagged (`in_scope = false`) rather than silently dropped, so every exclusion remains auditable. Each report's ATA chapter is derived from its two-digit JASC/ATA code and joined to a reference table built from the standard ATA-100 chapter nomenclature.

Volume & concentration (Pareto)

Reports are counted per ATA chapter and ranked by volume, with a running cumulative share — the standard 80/20 (Pareto) view used to identify where a small number of categories account for most of the activity.

Normalized rate

To compare families fairly (e.g. A320 vs. A330), raw counts are converted to a rate per 1,000 distinct aircraft observed. This is a **relative comparison indicator, not a true failure rate** — see the reporting-bias assumption above.

Age-band over-representation (lift)

For each ATA chapter × age band, a **lift** score is computed: observed report count divided by the count that would be expected if age had no relationship to chapter at all (`lift = observed / expected`). Lift > 1 means a chapter is over-represented in that age band; lift < 1 means under-represented. This is an association measure, not a statistical test of causation.

Text mining

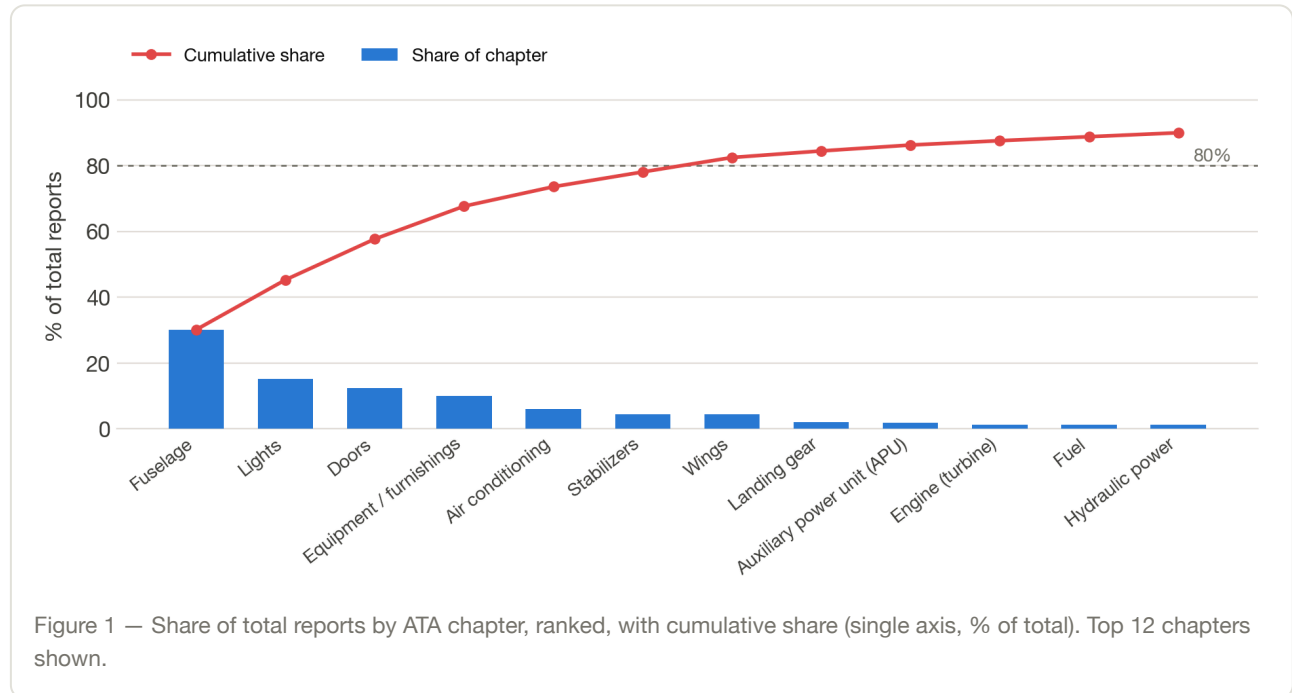
Free-text fault narratives (`Discrepancy` field, English) were cleaned (lowercased, numeric/part-number tokens stripped) and scored with **TF-IDF** (unigrams and bigrams, English stop-words removed) to surface terms that are distinctive to each ATA chapter, rather than just frequent overall. Topic modeling (e.g. LDA) was deliberately not used — TF-IDF keeps every term traceable to a defensible statistic rather than a modeled, harder-to-explain "topic."

Temporal trend

Annual report counts are shown both raw and normalized by the same fleet-size proxy used above, specifically to separate a genuine reliability trend from a fleet-growth artifact. 2026 is excluded from all year-over-year comparisons (partial year).

Where Maintenance Events Concentrate

Report volume is heavily concentrated: **Fuselage alone accounts for 30%** of all reports across the three families, and **seven chapters account for 82%** of total volume — close to a textbook 80/20 split.



Raw counts answer a *workload* question — where does maintenance effort concentrate today. A second, different question is which systems are *disproportionately* reported once fleet size is accounted for. The heatmap below shows reports per 1,000 aircraft observed, by family, for the same top chapters:

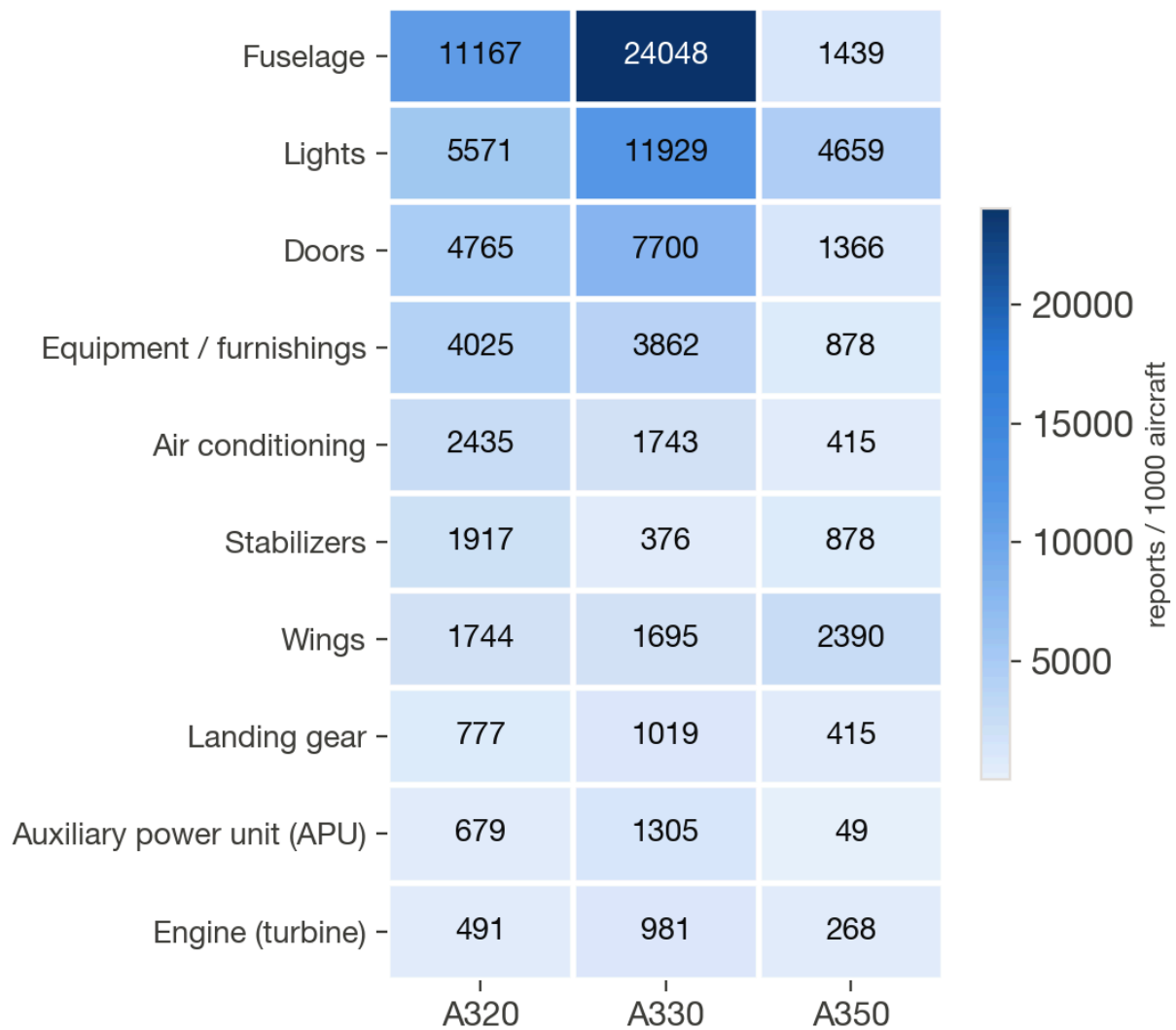


Figure 2 — Normalized report rate (per 1,000 aircraft observed) by ATA chapter and aircraft family. A330 shows a markedly higher normalized Fuselage and Lights rate than A320, a pattern invisible in raw counts (where A320 dominates simply because there are far more A320s in service).

Reading caution

The A330 (210 aircraft) and especially A350 (41 aircraft) normalized rates rest on much smaller samples than the A320 (2,568 aircraft) — they are more sensitive to statistical noise and should be treated as directional, not precise.

Age & Wear Patterns

Splitting reports into age bands (young / mid-life / aged, computed per family — see Section 05) reveals two clearly distinct profiles:

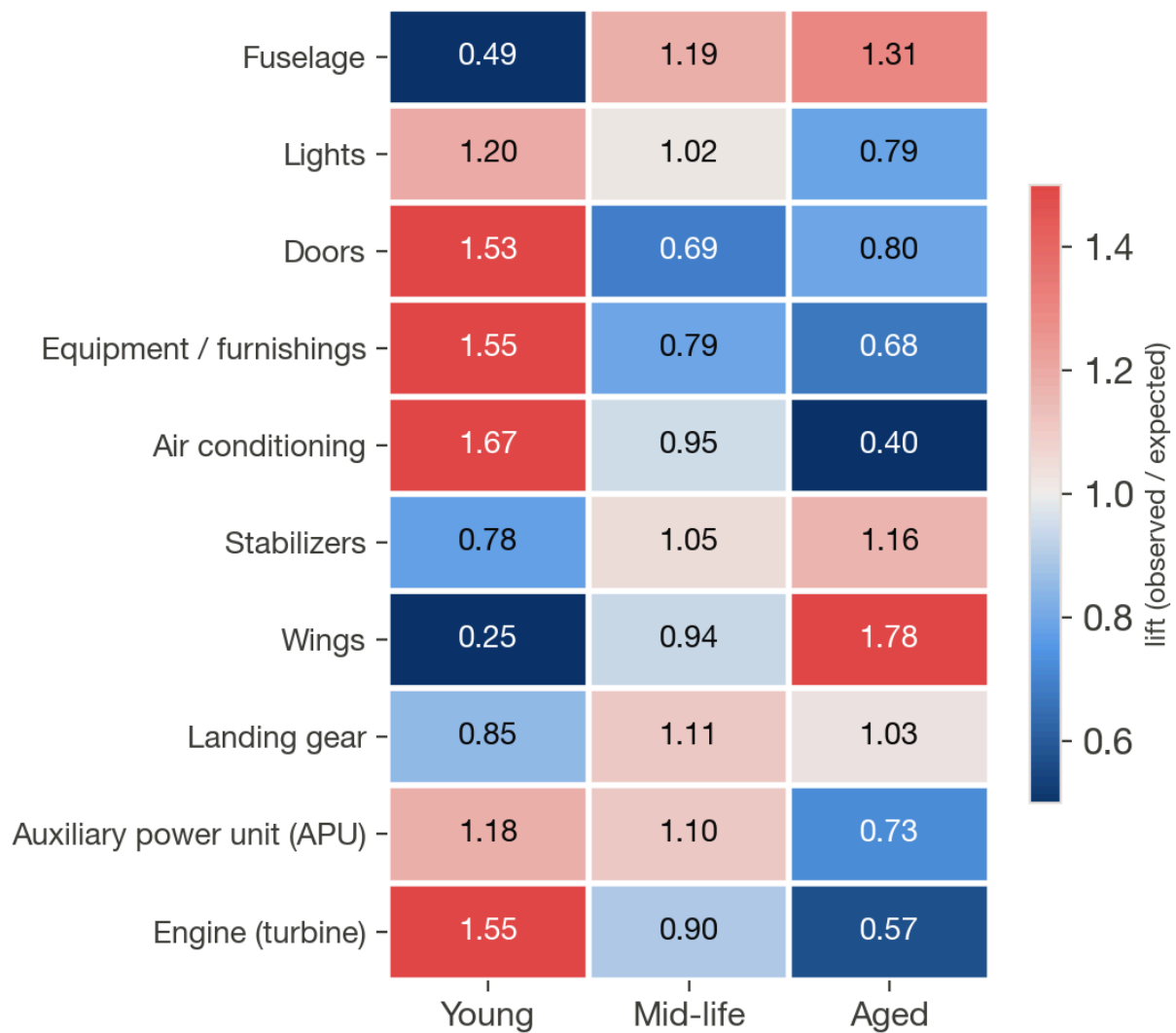


Figure 3 — Lift (observed ÷ expected reports) by ATA chapter and age band. Red = over-represented, blue = under-represented, white ≈ no age relationship.

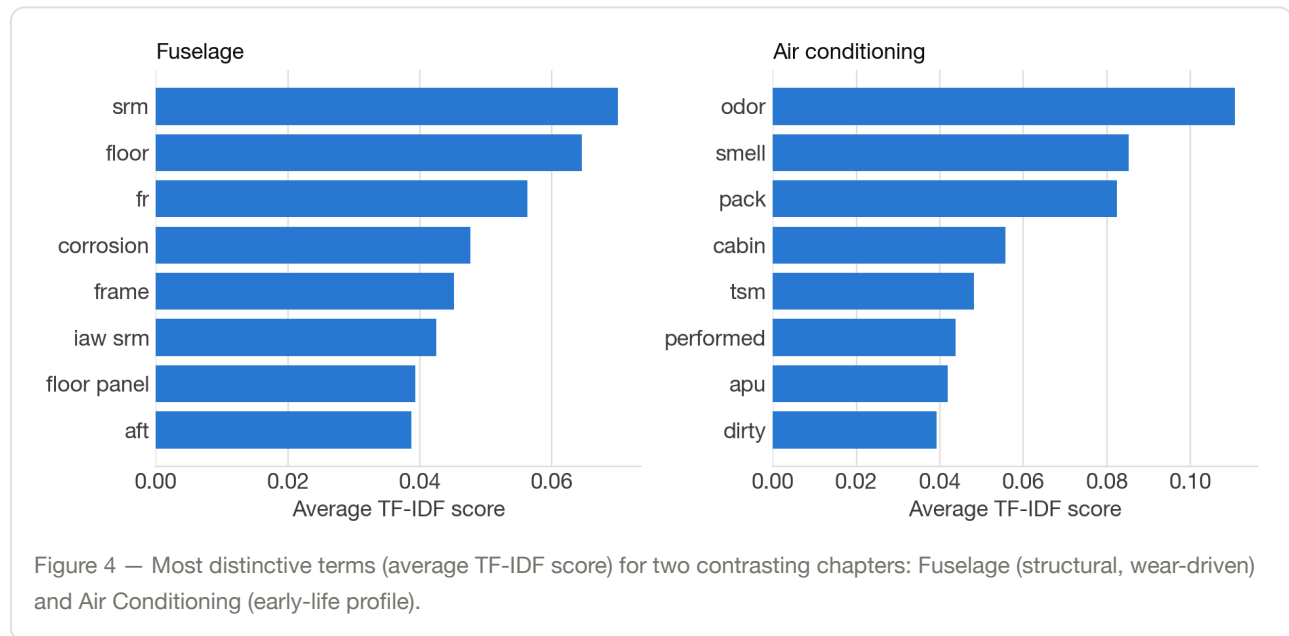
- **Wear / fatigue profile** — Wings (lift 1.78 on aged aircraft, 0.25 on young) and Fuselage (1.31 aged vs. 0.49 young) are increasingly reported as aircraft age. This is the signature of structural fatigue accumulating with flight cycles.
- **Early-life defect profile** — Air Conditioning (1.67 young vs. 0.40 aged), Doors, Equipment/Furnishings and Engine are all more reported on *younger* aircraft — consistent with entry-into-service issues that get resolved as fleets mature.

Why this distinction matters operationally

These two profiles call for different maintenance strategies: fatigue-driven systems benefit from inspection intervals that scale with cycles/age, while early-life systems benefit from tightened entry-into-service quality processes and supplier follow-up — not the same lever applied to both.

What the Technical Narratives Reveal

The free-text fault descriptions confirm — and explain — the age pattern above. TF-IDF surfaces the most distinctive vocabulary per ATA chapter:



- **"Corrosion" is a top term for both Fuselage and Wings** — the two chapters flagged as wear-driven in Section 07. This is a strong, independent confirmation from a completely different data field (free text vs. structured age statistics) of the same underlying phenomenon.
- **Air Conditioning is dominated by "odor" and "smell," not mechanical-failure vocabulary.** This is the most actionable and least expected finding in the dataset: the most frequent complaint in this system is a perceived cabin odor, not a classic hardware failure — pointing toward a different root-cause investigation (bleed air / pack contamination) than a standard parts-failure review.
- Other chapters show coherent, domain-specific vocabulary (e.g. hinge/elevator/stabilizer for Stabilizers, gear/brake/wheel for Landing Gear) — a useful sanity check that the text signal is real and not noise.

Is the Fleet Getting Less Reliable?

Raw report volume rose **×2.5 between 2015 and 2025** — a number that, read alone, looks like a reliability problem. Once normalized by the same fleet-size proxy (which itself grew **×1.9** over the same period), the picture changes substantially:

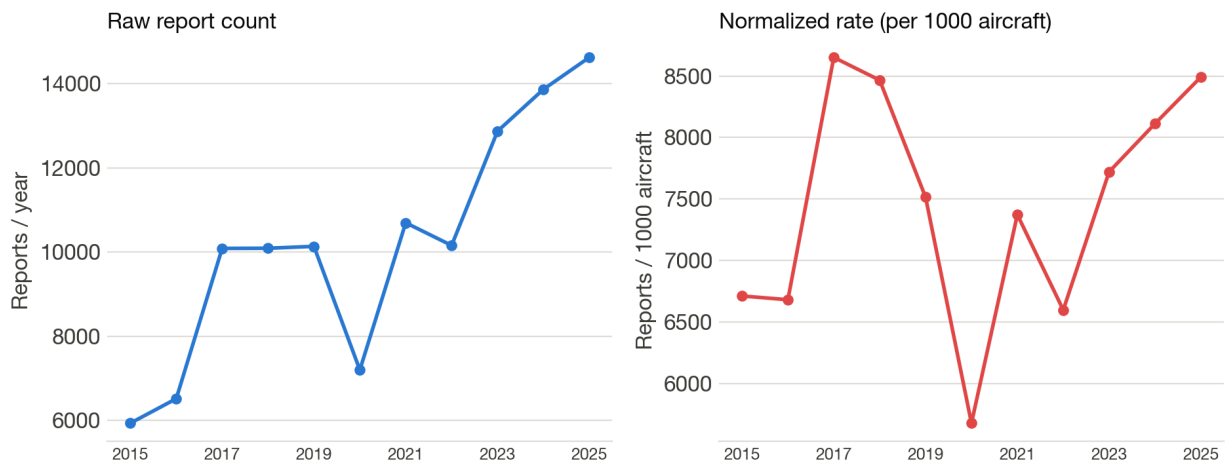


Figure 5 — Report volume 2015–2025, raw (left) vs. normalized per 1,000 aircraft (right). Same underlying data, different conclusion.

- **Most of the raw increase is explained by fleet growth**, not a deteriorating fleet. The normalized rate has no clear multi-year trend.
- **2020 stands out as a clear low point** in the normalized rate — consistent with a COVID-19 effect: fewer aircraft in active operation means fewer opportunities to detect and report a fault. A lower rate here reflects reduced exposure, not necessarily fewer underlying problems.

Reading caution

Single-year-over-year swings for an individual chapter can be noisy; a one-point comparison (e.g. 2024→2025) should not be read as a structural trend without confirmation over several years.

Limitations

Limitation	Implication
US-only geographic coverage	Results are a proxy for the US-operated Airbus fleet, not the global fleet. Regional maintenance conditions, climates, and utilization patterns elsewhere are not captured.
Event-reporting bias	Counts reflect what was reported, not a full census of what occurred. Fleet-size proxy under-counts true fleet size.
Small A350 sample	Only 41 distinct aircraft (2 in 2017, ramping to 33 in 2025). Any A350 comparison, especially in early years, carries wide uncertainty.
Undocumented fault-nature codes	NatureOfConditionA has no public FAA definition; retained in the data but not used as an analytical basis.
ATA chapter reference table	40 of 46 chapters confirmed with high confidence against standard ATA-100 nomenclature; 1 chapter (14) could not be documented with certainty.
Lift is an association, not a cause	Age-band over-representation shows correlation with aircraft age, not a proven causal mechanism — consistent with engineering expectations (fatigue, EIS defects) but not a substitute for engineering root-cause analysis.
Outlier handling	A small number of implausible cycle/hour values (8 cycle values, 139 hour values out of 140,139 rows) were nulled rather than capped, to avoid injecting false "old aircraft" signal.

Recommended Actions

Each recommendation below is tied directly to a finding in Section 06–09 — the goal is that this analysis feeds a decision, not just a chart.

1 Prioritize the seven high-volume ATA chapters in maintenance program reviews

VOLUME / WORKLOAD

Fuselage, Lights, Doors, Equipment/Furnishings, Air Conditioning, Stabilizers and Wings drive 82% of unscheduled maintenance volume. A focused review of tooling, spares stocking, and technician training for these seven systems will have outsized impact relative to spreading effort evenly across all 46 chapters.

2 Scale structural inspection intervals to cycles for Fuselage and Wings

AGE & WEAR

Both chapters show a strong, cycle-linked over-representation on aged aircraft, corroborated independently by "corrosion" as a top narrative term. This supports reinforcing (or validating existing) corrosion-prevention and structural inspection programs that scale with flight cycles rather than calendar time alone.

3 Open a dedicated root-cause investigation into cabin odor events

TEXT MINING

Air Conditioning reports are dominated by "odor"/"smell," not classic mechanical failure terms. This is a different problem category than a parts failure — likely warranting a bleed-air/pack contamination investigation and a distinct triage path from standard AC system troubleshooting.

4 Tighten entry-into-service quality follow-up for early-life-flagged systems

AGE & WEAR

Doors, Equipment/Furnishings and Engine chapters are more reported on younger aircraft — a classic EIS defect signature. Supplier quality escalation and post-delivery inspection follow-up on these systems, concentrated in an aircraft's first years in service, should reduce this bucket faster than age alone will.

5**Report maintenance-volume trends using the normalized rate, not raw counts****TREND / REPORTING**

Raw counts ($\times 2.5$ over 10 years) overstate a reliability signal that is mostly fleet growth ($\times 1.9$). Internal and leadership reporting on maintenance trend should lead with the per-1,000-aircraft rate to avoid a false "the fleet is degrading" narrative.

6**Treat A330 and A350 normalized comparisons as directional, not conclusive****DATA QUALITY**

With 210 and 41 distinct aircraft respectively, normalized rates for these families are more volatile than the A320's. Any resourcing decision based on an A330/A350 rate comparison should be corroborated with a larger internal dataset before being finalized.

Conclusion & Next Steps

Across four independent analyses — volume mapping, age-based over-representation, text mining, and temporal trend — the same story reinforces itself from different angles: a small set of systems (Fuselage and Wings) show a genuine, cycle-linked wear signature corroborated by both statistics and technician language, a distinct set of systems (Air Conditioning, Doors, Engine) show an early-life defect signature, and the apparent overall upward trend in maintenance volume is largely explained by fleet growth rather than declining reliability. That convergence across independently-computed metrics is what gives these findings enough confidence to act on.

Suggested extensions

- **Apply this method to internal maintenance data.** The same four analyses (Pareto, normalized rate, age-lift, TF-IDF) could run directly on an airline's or MRO's internal fault records, removing the US-only geographic bias and the event-reporting undercount entirely.
- **Seasonality.** A time-of-year breakdown (e.g. de-icing-related systems in winter months) was out of scope here but is a natural next cut of the same data.
- **Operator-level comparison.** Comparing report patterns across operators could surface useful signal, but needs careful framing to avoid unfair operator-to-operator comparisons from reporting-behavior differences alone.
- **Topic modeling as an enrichment layer.** TF-IDF was deliberately chosen for defensibility; a topic model (LDA/NMF) could be added later as a complementary, more exploratory layer once the TF-IDF baseline is validated against domain expertise.

Full detail

The complete analysis — reproducible notebooks, source code, and an interactive filterable dashboard — is available at the project repository and the live Streamlit application referenced on the cover page.